

Nonlinear Computing with Optical Twinning Trees

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A twinning field is one electromagnetic drive under which two distinct materials produce the same optical response. It was originally calculated in advance from models of both systems. Here we show that the field can instead be found by feedback: a balanced detector measures the response difference, and an integral controller changes the field incident on both systems until that difference vanishes. The equality is then a physical root condition. We derive the stability of this loop, distinguishing the familiar susceptibility approximation from the full cavity-controller dynamics, and show that a changing root is followed with a finite lag. Exact propagation of two four-site Hubbard rings verifies the predicted tracking error. We then reverse the construction. The coefficients of a quadratic set the parameters of a Kerr cavity and a linear reference cavity, whose non-vacuum equal-intensity states encode the accessible positive roots. Several quadratic conditions can share the same reference, and their error signals can be multiplied to form an optical twinning tree. Full cavity simulations find four roots with two quadratic branches and six roots with three. With three branches the local convergence rates differ by a factor of 117.5, and small cavity-parameter errors can preserve an optical lock after the intended root has been lost. The result is a nonlinear optical root finder with an explicit stability and conditioning theory. Its experimental realisation remains to be demonstrated.

I TWINNING FIELDS IN NONLINEAR OPTICS

A material is often introduced optically by a list of things it does to light. That shorthand is powerful because the weak-field answer is reproducible. Under strong driving, however, the interrogating field becomes a participant: it can redirect a many-body current [1], force a chosen observable trajectory [2], or suppress one component of a composite response [3]. The map from field to nonlinear response need not even be one-to-one [4]. At that point the familiar division between a property of the material and a property of the experiment begins to soften. If the field helps to make the answer, how much of a material's optical identity belongs to the material alone?

Twinning fields give one answer. A twinning field is a single electromagnetic drive $\Phi_\tau(t)$ for which two distinct systems return the same selected response,

$$J_1[\Phi_\tau](t) = J_2[\Phi_\tau](t).$$

The construction was introduced for nonlinear many-body systems and demonstrated for Fermi-Hubbard and tight-binding materials [5]. Its insistence on one identical field is the point. With two independently tailored fields, tracking control can make different systems reproduce the same target by correcting each one separately [1, 2]. A twinning field instead identifies a trajectory on which two distinct response laws coincide under one shared physical cause.

In the original formulation, that trajectory is calculated in open loop from the evolving states of both systems [5]. This establishes the object, but leaves an experimental loop to close. The field must be known before it is applied, while the information used to calculate it is returned by the responses produced after application. Feedback resolves the apparent circle by allowing

the materials to supply that information as the experiment proceeds. Tracking feedback can generate a prescribed broadband optical behaviour [6], and a measured response mismatch can be amplified and returned without first shaping a correction pulse [7]. Those architectures regulate one response or correct one branch towards another. The stricter task here is to adapt the one field incident on both systems.

Once the problem is written this way, a second interpretation becomes difficult to avoid. Balanced detection measures $J_1 - J_2$. Laser locking calls such a signed difference an error signal [8, 9]. Numerical analysis calls it a residual. A controller that adjusts Φ until $J_1 - J_2 = 0$ is therefore finding a root, with the two physical systems evaluating the function. The shift is physical as well as linguistic: the field that interrogates the materials becomes the unknown, and their disagreement supplies each iteration. There is already a substantial optical-computing world around this observation. Designed electromagnetic media perform mathematical operators [10] and solve integral equations [11]. Coherent feedback supplies analogue solutions of partial differential equations [12], microring networks solve ordinary differential equations [13], and inverse-designed free-space feedback solves integral equations [14]. Delayed and integrated photonic reservoirs process information in their dynamics [15, 16], while optoelectronic recurrence searches for computational fixed points [17, 18]. The question here is consequently narrow: can equality between distinct systems under one field encode an equation, and can feedback find its solutions?

Before using the loop as a computer, we must establish what it can follow. The derivative of the response difference gives the restoring feedback sign and, when the optical response is fast, the convergence rate. It is not by itself the stability analysis of a dynamical cavity.

Nor can an integral controller follow a changing equality while maintaining exactly zero error, since zero error would stop the controller. We derive both qualifications and test the second by propagating two four-site Hubbard rings. The common phase follows the field that would instantaneously equalise their currents, but with a lag fixed by the speed of that field and the feedback gain.

We then ask a different question: can the equality be chosen so that its solutions are the roots of a supplied equation? Nonlinear optical response can serve as a computational transformation [19], and Hamiltonian parameters can encode data for a physical processor [20]. We use the coefficients of a quadratic to set the detuning and linewidth of a Kerr cavity relative to a linear reference. Except for the trivial equality between two empty cavities, equal output intensities then occur at the accessible positive roots. Several Kerr cavities can share the same reference, and multiplication of their error signals combines the corresponding zero sets. Optical multiplication, layered nonlinear response, and programmable photonic transformations provide the required neighbouring operations [21–23]. Full cavity simulations recover four roots from two quadratic factors and six roots from three, using separately initialised runs with the appropriate restoring feedback sign.

The hierarchy also exposes its own limits. With three quadratic factors the fastest local convergence rate is 117.5 times the slowest. Detector dynamics and delay restrict the stable feedback range, while cavity-parameter errors can shift the optical equality away from the root encoded by the nominal coefficients. A loop may therefore remain locked and still return the wrong answer. These are numerical results for small model systems, not an experimental demonstration or a claim of optical advantage. Section II defines the common-field feedback problem. Section III derives its stability and tracking laws, which Sec. IV tests for the Hubbard pair. Section V encodes a quadratic in two cavities. Sections VI and VII compose several such conditions and determine the resulting limits. We return to the computational interpretation in Sec. VIII.

II A TWINNING FIELD AS A PHYSICAL ROOT

Consider two systems with internal states $x_1(t)$ and $x_2(t)$. Their response to a drive depends both on its present value and on the state prepared by its history, which we write as $J_j[\Phi(t), x_j(t)]$. We define strict twinning by requiring the same total field $\Phi(t)$ to reach both systems and the chosen responses to agree over an observation interval $\mathcal{T}_{\text{obs}}$,

$$J_1[\Phi(t), x_1(t)] = J_2[\Phi(t), x_2(t)], \quad t \in \mathcal{T}_{\text{obs}}.$$

This is stronger than asking one branch to imitate the other using an additional correction field. The equality concerns what the two systems do under one shared physical cause [5].

The quantity available to a balanced detector is the differential response

$$e(t) = J_1[\Phi(t), x_1(t)] - J_2[\Phi(t), x_2(t)]. \quad (1)$$

Equation (1) is simultaneously the optical error signal and the residual of the root problem. This identification carries more than a change of vocabulary. The unknown Φ is not passed to a numerical routine together with analytic functions J_1 and J_2 . It is applied to the systems, and their measured difference evaluates e . Signed optical zero crossings are the ordinary currency of cavity locking [8, 9]. Here the two arms contain different materials, and equality between their responses, rather than one cavity resonance, defines the lock point.

At time t , the controller has access only to measurements that have returned by t and to its own stored state. With a total propagation, detection, and processing delay $\tau > 0$, it cannot know the response to the field it is applying at that same instant. Equality from an arbitrary initial time is therefore unavailable to a causal feedback loop. The controller can instead acquire a root over an interval $\mathcal{T}_{\text{acq}}$ and then test the resulting field over $\mathcal{T}_{\text{obs}}$. Delayed response information is a standard ingredient of nonlinear control [24] and of computational feedback dynamics [15]. Related optical response-control schemes use it directly [6, 7]. Here the update acts before the passive split, so both systems continue to receive exactly the same field.

The simplest controller that retains a nonzero field when the error vanishes is an integrator,

$$\dot{\Phi}(t) = -\kappa\sigma e(t - \tau), \quad (2)$$

where $\kappa > 0$ is the gain and $\sigma = \pm 1$ fixes the feedback polarity. A proportional correction would disappear at $e = 0$ and return the apparatus to its uncorrected drive. By contrast, Eq. (2) stores the acquired value in the controller state. At an interior stationary point,

$$\dot{\Phi} = 0 \iff e(\Phi_\star) = 0,$$

provided the two systems have settled into the corresponding driven states. A regular twinning field is thus a fixed point of the loop, and a stable fixed point is a twinning field for the selected response [2]. This equivalence holds for a stationary equality. If the equality selected by the evolving states moves, then exact zero error also makes the integrator stop. Following such a root requires a finite mismatch or an additional model of its motion. Section III derives the resulting tracking law.

This equivalence also fixes what we mean by success. During $\mathcal{T}_{\text{acq}}$, returned measurements are permitted to change Φ . Once a residual threshold has been maintained for a prescribed hold time, the controller state is retained and the equal-response claim is assessed on the subsequent observation interval. A transient crossing of $e = 0$ is not a solution, and equality during acquisition is not asserted before information has had time to circulate.

Figure 1 puts this distinction into an optical layout. A common modulator prepares Φ , a passive splitter sends it to the two systems, and balanced detection forms Eq. (1). The signed integrator implements Eq. (2). Every component has a close optical precedent in cavity locking and feedback computing [9, 14]. The modulator is placed before the splitter so that no system receives an independent correction. Open-loop synthesis, asymmetric correction, and common-field acquisition therefore differ in where the controllable field enters the apparatus. We next determine when the resulting feedback is stable and how closely it can follow an equality that changes in time.

III STABILITY AND TRACKING

A stable lock requires the response difference to reveal which way the field should move. Let Φ_* denote a stationary twinning field, and first suppose that the material states do not change while the field is perturbed. The relevant slope is

$$\chi_\Delta = \partial_\Phi[J_1 - J_2]|_{\Phi_*}. \quad (3)$$

Its sign fixes the feedback polarity. Its magnitude fixes the rate at which a small displacement from Φ_* changes the measured error.

Writing $\Phi = \Phi_* + \delta\Phi$, Eq. (2) becomes

$$\dot{\delta\Phi}(t) = -\kappa|\chi_\Delta|\delta\Phi(t - \tau).$$

The characteristic equation is [25]

$$\lambda + \kappa|\chi_\Delta|e^{-\lambda\tau} = 0, \quad \kappa|\chi_\Delta|\tau < \frac{\pi}{2} \quad (4)$$

on the branch connected to the stable zero-delay solution. A numerical sweep of the nonlinear scalar equation places the loss of stability at $\kappa|\chi_\Delta|\tau = 1.5675$, within 0.0033 of $\pi/2$. This calculation assumes that the response follows the field instantaneously. It gives neither the optical eigenmodes of a cavity nor the error incurred when the equality itself changes.

The latter error follows directly. Let $\Phi_*(t)$ solve $e[\Phi_*(t), t] = 0$, and write $\delta(t) = \Phi(t) - \Phi_*(t)$. With zero delay,

$$\dot{\delta} = -\kappa|\chi_\Delta(t)|\delta - \dot{\Phi}_*(t) + O(\delta^2). \quad (5)$$

For a slowly changing regular root,

$$\delta(t) \simeq -\frac{\dot{\Phi}_*(t)}{\kappa|\chi_\Delta(t)|}.$$

An integral controller therefore needs a nonzero error to keep moving. The error grows with the speed of the twinning field and decreases with gain, until the gain itself threatens stability. Cancelling the leading term requires an estimate of $\dot{\Phi}_*$, which is information not contained in the instantaneous error alone.

The cavity problem has a second timescale because the optical amplitudes do not respond instantaneously. Let x contain their real quadratures and let s denote the common input power:

$$\dot{x} = f(x, s), \quad \dot{s} = -g\sigma E(x).$$

At a stationary equality, define $A = \partial_x f$, $b = \partial_s f$, and $c = \partial_x E$. Small perturbations are governed by

$$J_{\text{loop}} = \begin{pmatrix} A & b \\ -g\sigma c^\top & 0 \end{pmatrix}. \quad (6)$$

If the optical modes are stable and relax much faster than the controller, then $E'(s_*) = -c^\top A^{-1}b$ and the slow eigenvalue is

$$\lambda_{\text{slow}} = -g\sigma E'(s_*) + O(g^2/\gamma_{\text{opt}}).$$

The slope of the steady response is thus the low-gain limit of the full stability calculation. Outside that limit the eigenvalues of Eq. (6), not the slope alone, decide whether the lock is stable.

Figure 2 makes the distinction explicit for the cavity systems used below. At low gain, the slope approximation differs from the dominant decay rate by at most 1.06% for the quadratic pair and 1.26% for the two-factor construction. Increasing the gain eventually makes optical relaxation the limiting timescale. The lower quadratic root becomes unstable at $g = 15.77$, while the best-conditioned root of the two-factor system becomes unstable at $g = 1.10$. The remaining roots do not cross into instability within their recorded gain ranges. Equation (4) is retained only for the reduced delayed equation. Delay in the full nonlinear cavity model is examined numerically in Sec. VII.

We have therefore identified two independent departures from an ideal scalar root finder. Finite response time changes the stability boundary. Motion of the equality produces a tracking error even when the stationary loop is stable. The next section tests the second effect in the interacting many-body system for which twinning fields were originally proposed.

IV TWINNING TWO HUBBARD RINGS

We now return to the many-body setting of the original twinning-field problem. The two systems are half-filled four-site spinful Fermi–Hubbard rings,

$$H_j(\Phi) = -t_0 \sum_{\ell, \varsigma} \left(e^{-i\Phi} c_{\ell\varsigma}^\dagger c_{\ell+1, \varsigma} + \text{H.c.} \right) + U_j \sum_{\ell} n_{\ell\uparrow} n_{\ell\downarrow}. \quad (7)$$

We take $U_1 = 0.5$ and $U_2 = 2.0$. The interactions are different, but both rings receive the same Peierls phase $\Phi(t)$. Current control in driven many-body systems supplies the response channel [1, 2], while the requirement of one common phase is inherited from the twinning construction [5].

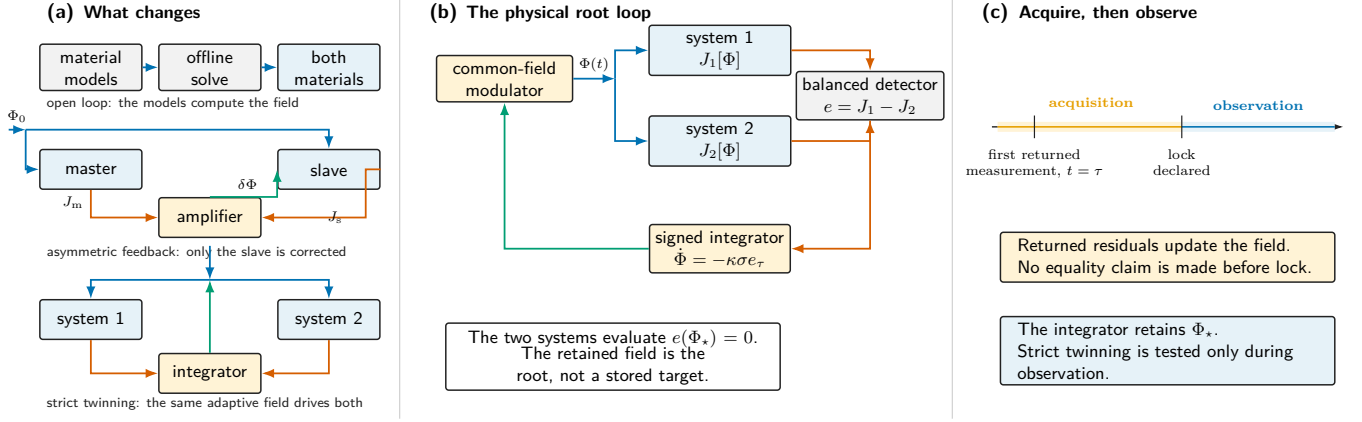


FIG. 1. The common field feedback architecture illustrates how a twinning field becomes a physical root. (a) Open-loop synthesis solves for a field in advance, asymmetric master–slave feedback corrects one branch, and strict twinning instead sends one identical total field to both systems. (b) The two systems evaluate the differential response $e = J_1 - J_2$, a balanced detector measures it, and a signed integrator updates the shared field. (c) Nonzero latency introduces an acquisition interval before the observation window on which strict twinning is asserted. The diagram specifies a proposed architecture rather than an experimental realisation.

The current in each ring is

$$\hat{J}_j(\Phi) = -\partial_\Phi H_j(\Phi), \quad J_j(t) = \langle \psi_j(t) | \hat{J}_j[\Phi(t)] | \psi_j(t) \rangle,$$

and the two states evolve independently under their respective Hamiltonians. Fixed particle number and spin reduce each ring to a 36-dimensional sector. The controller sees only the difference $J_1 - J_2$. It is not given either wavefunction or either interaction strength.

To measure the tracking error, we also solve $J_1(\Phi, t) = J_2(\Phi, t)$ from the two propagated states at each time. This independent calculation defines the instantaneous twinning field $\Phi_*(t)$, but its value is withheld from the feedback loop. The comparison is important because the root moves as the states evolve. Equation (5) identifies the difference $\Phi - \Phi_*$ as the error that allows a pure integrator to continue changing the applied phase.

We vary the speed of the material evolution relative to the controller, the gain, and the delay. In all 18 stable integral-feedback runs, the root-mean-square phase error increases with the speed of Φ_* and decreases with gain, as Fig. 3(c) shows. For the representative run with $g = 36$ and $\tau = 0.02$, the measured RMS error is 0.01753. The leading approximation $|\dot{\Phi}_*|/(g|\chi_\Delta|)$ gives 0.01729. The associated tail RMS current difference is 0.02571.

We also make a state-based comparison by adding the independently calculated $\dot{\Phi}_*$ to the feedback law. This feed-forward term reduces the representative RMS phase error to 4.55×10^{-4} and the current difference to 4.83×10^{-4} . Across the sampled parameters it removes between 91% and 99% of the phase error. It does so by using the internal states to reconstruct the quantity that the detector alone cannot provide. We therefore use it to verify the missing term in Eq. (5), not as the proposed experimental controller.

The timestep refinement in Fig. 3(d) confirms that the

two curves have different limits. Halving the timestep from 0.004 to 0.002 changes the integral-feedback error by 2.90%, consistent with convergence to a finite tracking lag. The remaining error in the state-based calculation falls approximately linearly with the timestep, as expected from its first-order estimate of $\dot{\Phi}_*$. These small-system calculations do not supply a general many-body stability theorem. They show instead that two interacting systems can determine a common twinning field by feedback, and that following a changing field without access to their states necessarily incurs the lag derived in Eq. (5).

V ENCODING A QUADRATIC IN TWO CAVITIES

The preceding sections find a field for an equality fixed by two given materials. We now reverse the problem. Suppose that the desired values are the roots of

$$q(n) = n^2 + bn + c.$$

We seek two optical response curves that cross precisely at those values. A Kerr cavity is useful because its steady input–output relation is cubic in the intracavity intensity. Such cavities exhibit optical bistability [26], admit a quantum input–output description [27], and can support several stationary branches [28, 29]. Here the nonlinearity is chosen so that the difference between two cavity responses contains the supplied quadratic.

For a coherently driven single-mode cavity,

$$\begin{aligned} \dot{a}_j &= [-\gamma_j + i(\Delta_j - K_j|a_j|^2)] a_j + \eta, \\ P_j(n) &= n[\gamma_j^2 + (\Delta_j - K_j n)^2] = |\eta|^2, \end{aligned} \quad (8)$$

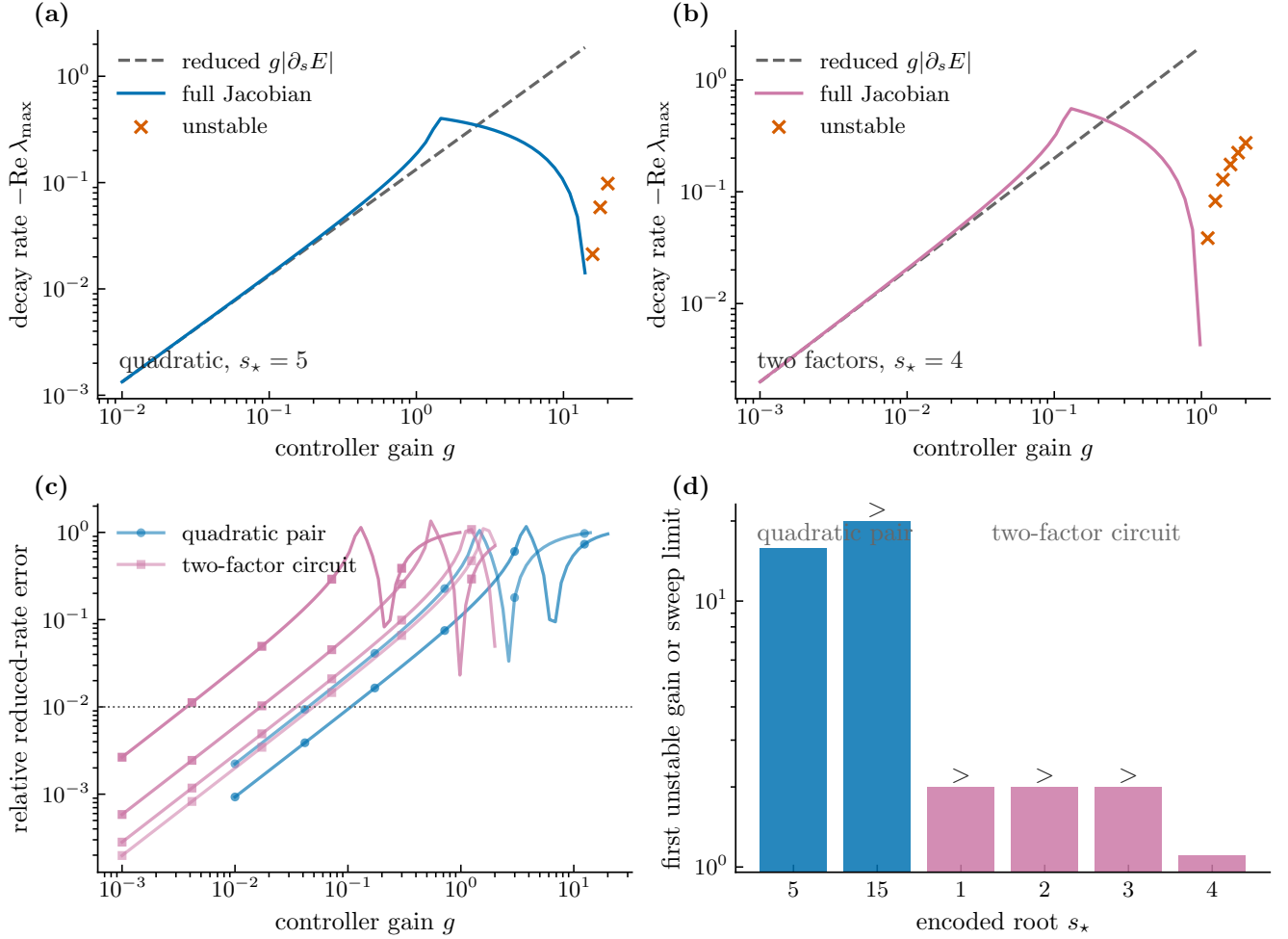


FIG. 2. The steady-response slope gives the low-gain limit of the full cavity-controller spectrum. (a) At the lower root of the quadratic encoding, the dominant eigenvalue of the five-variable Jacobian follows the slope prediction at small gain, departs when optical relaxation becomes rate limiting, and then crosses into instability. (b) The same comparison for the best-conditioned root of the two-factor circuit. Crosses mark unstable gains. (c) Relative error of the slope prediction at every encoded root. It remains below 1.06% for the quadratic pair and 1.26% for the two-factor circuit over the low-gain ranges used in the comparison. (d) First unstable gain, or the largest tested stable gain where no crossing is observed. These spectra have zero delay. Equation (4) applies to the reduced delayed equation.

where $n = |a_j|^2$, γ_j is the linewidth, Δ_j the detuning, and K_j the Kerr coefficient. The function $P_j(n)$ is the input power required to sustain the intensity n on a chosen steady branch. Two cavities driven by the same field have the same intensity wherever $P_1(n) = P_2(n)$.

Let cavity 1 have $K_1 = 1$, and let cavity 2 be linear with $K_2 = 0$ and $\Delta_2 = 0$. Their difference is

$$P_1(n) - P_2(n) = n [(\gamma_1^2 + \Delta_1^2 - \gamma_2^2) - 2\Delta_1 n + n^2].$$

Matching the expression in brackets to $q(n)$ gives

$$\begin{aligned} K_1 &= 1, & K_2 &= 0, & \Delta_1 &= -b/2, & \Delta_2 &= 0, \\ \gamma_1^2 &= L, & \gamma_2^2 &= L + \Delta_1^2 - c, \\ P_1(n) - P_2(n) &= nq(n). \end{aligned} \quad (9)$$

The positive constant L keeps both linewidths physical

and can be chosen so that the relevant branches remain monotonic.

The factor n in Eq. (9) has a simple origin. At $n = 0$, both cavities are empty and are therefore indistinguishable for any quadratic. This universal vacuum equality does not encode a root. On the non-vacuum part of the chosen branches,

$$P_1(n) = P_2(n) \iff q(n) = 0, \quad n > 0.$$

Each accessible positive root r is thus represented by an equal-intensity state, reached at the common input power

$$s_* = P_1(r) = P_2(r).$$

The coefficients set the cavity parameters, the measured intensity difference evaluates the equation, and the power

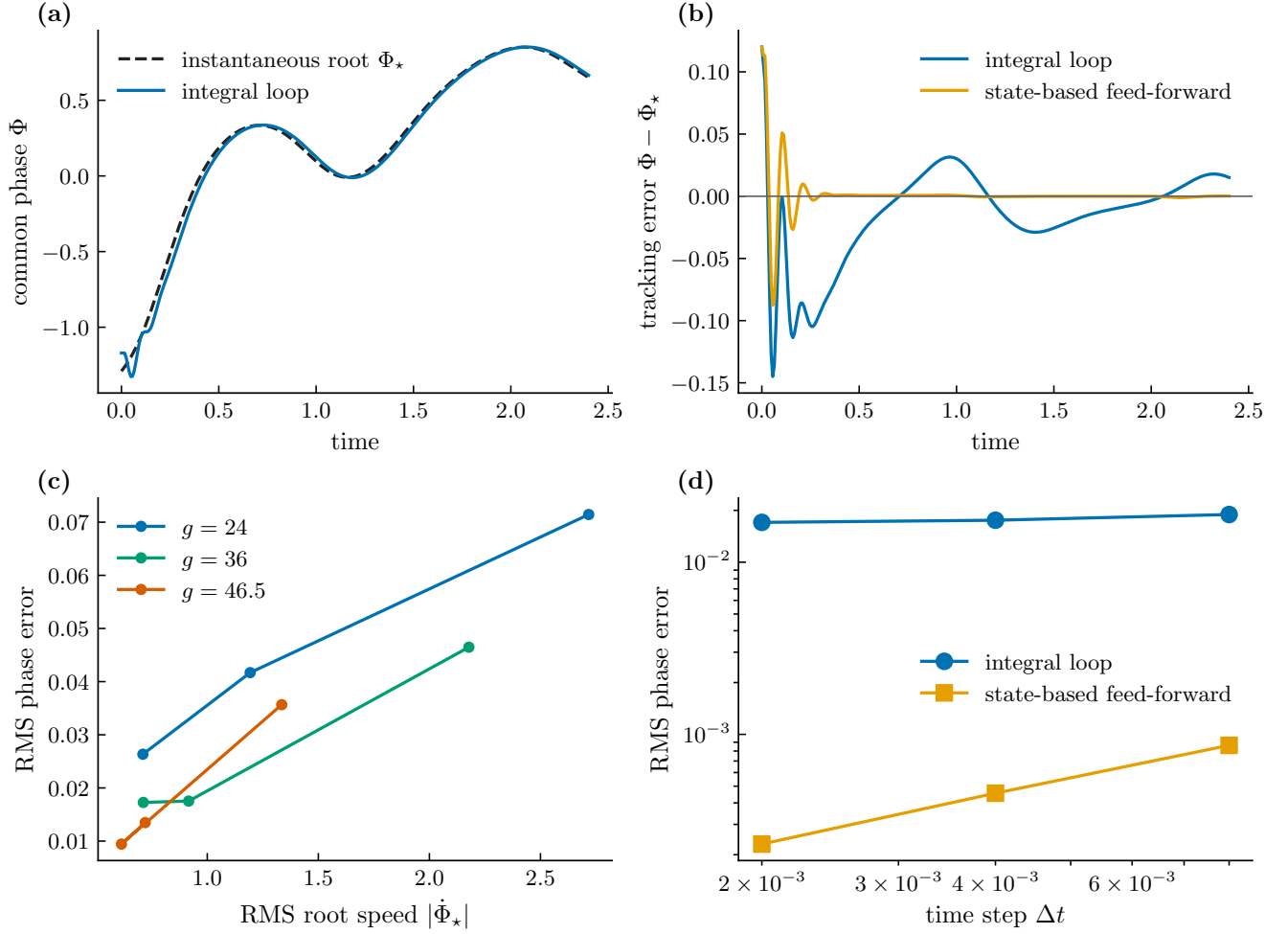


FIG. 3. Integral feedback follows a changing Hubbard twinning field with finite lag. (a) The common Peierls phase obtained from the current difference is compared with the equality independently calculated from the two four-site states. (b) The feedback error remains finite while that equality moves. A state-based feed-forward comparison removes the leading lag, but requires the internal states and is not the proposed detector-only controller. (c) The RMS phase error increases with the speed of the twinning field and decreases with gain in the stable region. (d) Timestep refinement approaches a finite error for integral feedback and zero error for the state-based comparison.

retained by the controller identifies the root. This is a specific form of parameter encoding in a physical processor [20, 23].

During acquisition we integrate the full complex amplitudes rather than replacing the cavities by their steady response curves. If $n_j(s, t)$ is the returned intensity under common power s , then

$$\dot{s} = -g\sigma[n_1(s, t) - n_2(s, t)], \quad \delta\dot{s} = -g\sigma e'(s_*)\delta s. \quad (10)$$

The second relation is obtained only after the cavity amplitudes have followed their steady branches. The full local stability is governed by the five-variable Jacobian in Eq. (6). The two roots of a quadratic usually have opposite slopes, so the feedback sign that restores one root destabilises the other. The sign selects a basin of attraction. It does not encode the value of the root.

For $q(n) = n^2 - 4n + 3 = (n-1)(n-3)$, Eq. (9) gives $P_1 - P_2 = n(n-1)(n-3)$. The two encoded roots occur at common powers $s = 5$ and $s = 15$. Full cavity simulations reach $n = 1$ with one feedback sign and $n = 3$ with the other. Fifteen of 22 runs, sampling initial power and both signs, reach one of these roots with a maximum intensity error of 2.18×10^{-6} . The other runs start outside the corresponding basin and encounter the imposed power limit. Each run reaches one attractor. The two-root solution is obtained by combining the successful runs.

Figure 4 follows the construction from the supplied polynomial to the measured intensity difference and then to the full dynamical trajectories. It also shows the finite range of the computation. In a separate sweep, 17 of 18 upper power limits below $s = 15$ exclude the higher root,

while all ten limits above it permit acquisition. The roots returned by the apparatus are therefore restricted to the monotonic cavity branches and power interval for which the encoding was designed. Unlike optical media that directly implement a mathematical operator [10, 11, 14] or a differential equation [12, 13], this construction evaluates an equation through equality between two nonlinear optical responses. Appendix B records the parameter bounds and numerical protocol.

VI COMPOSING QUADRATIC TWINNING CONDITIONS

One cavity pair represents one quadratic factor. A higher-order zero set can be assembled by making several such pairs depend on the same input power. If their measured intensity differences are $e_j(s)$, then

$$\mathcal{Z}\left(\prod_j e_j\right) = \bigcup_j \mathcal{Z}(e_j),$$

where $\mathcal{Z}$ denotes the zero set. The factorisation is supplied when the cavities are chosen. The optical system does not discover it.

To avoid duplicating the linear cavity, let several Kerr cavities share one reference with $P_0(n) = Gn$. For each supplied factor $q_j(s) = s^2 + b_js + c_j$, choose

$$\begin{aligned} \Delta_j &= -\frac{b_j K_j}{2G}, \\ \gamma_j^2 &= G - \Delta_j^2 + \frac{c_j K_j^2}{G^2}, \\ P_j(n) - P_0(n) &= \frac{n K_j^2}{G^2} q_j(Gn). \end{aligned} \quad (11)$$

Away from the common vacuum solution, branch j and the reference have equal intensity at the accessible positive roots of q_j .

The input–output functions in Eq. (11) design the cavities. The feedback signal is formed from their measured intensities:

$$\begin{aligned} e_j &= |a_j|^2 - |a_0|^2, \\ E_D &= \prod_{j=1}^D \frac{e_j}{\epsilon_j}, \quad \dot{s} = -g\sigma E_D. \end{aligned} \quad (12)$$

The constants ϵ_j put the detector signals on comparable scales. Analogue optical multiplication has been demonstrated at the component level [21], and repeated optical multiplication can build higher polynomial order [22]. Figure 5 combines these operations with one common drive and one reference cavity. We call the resulting hierarchy an *optical twinning tree*. Its leaves are the Kerr cavities, and the root of the diagram is the single feedback controller.

The first tree has two leaves,

$$q_1 = (s-1)(s-2), \quad q_2 = (s-3)(s-4).$$

Its feedback signal vanishes at the four roots of the supplied factorised quartic. The dynamical model contains three complex cavity amplitudes and one real controller variable. Of 42 runs sampling initial power and both feedback signs, 33 reach a root and nine reach the imposed power limit. The successful runs include all four roots, although each individual run reaches only one.

Adding

$$q_3 = (s-5)(s-6)$$

gives three Kerr leaves, one reference cavity, and six non-vacuum roots. We choose one gain at this depth and use the restoring feedback sign for each targeted root. All six roots are reached in the full nine-variable dynamics. Repeating each calculation with timesteps 0.02, 0.01, and 0.005 leaves every success classification unchanged. The largest difference in final power between the two finest steps is 5.71×10^{-6} .

The additional roots come with a conditioning penalty. At a simple root supplied by leaf k ,

$$E'_D(s_*) = \frac{e'_k(s_*)}{\epsilon_k} \prod_{j \neq k} \frac{e_j(s_*)}{\epsilon_j}. \quad (13)$$

Every extra leaf multiplies the local feedback slope by its off-root response. In the three-leaf example, the largest local rate is 117.5 times the smallest. The slowest root takes 1708.37 dimensionless cavity time units to meet the acquisition tolerance, compared with 9.19 for the fastest. Figure 6 shows the corresponding trajectories and settling times.

These calculations demonstrate a finite hierarchy of one, two, and three supplied quadratic factors. They do not show that the hierarchy scales favourably. Nor does one run enumerate the roots, discover the factorisation, or encode an arbitrary polynomial written in expanded form. Equation (13) explains the immediate difficulty: the same multiplication that combines zero sets can make the feedback slope at different roots extremely uneven. A larger optical twinning tree is therefore a larger equation only if its weakest branch still provides a measurable restoring signal.

VII CONDITIONING AND CALIBRATION

The feedback sign can select a root only when the measured difference changes sign there. For $q(n) = (n-r_-)(n-r_+)$,

$$q'(r_{\pm}) = \pm(r_+ - r_-).$$

The opposite signs distinguish the two roots. As the roots approach one another, however, the slope and hence the local information in the error signal vanish.

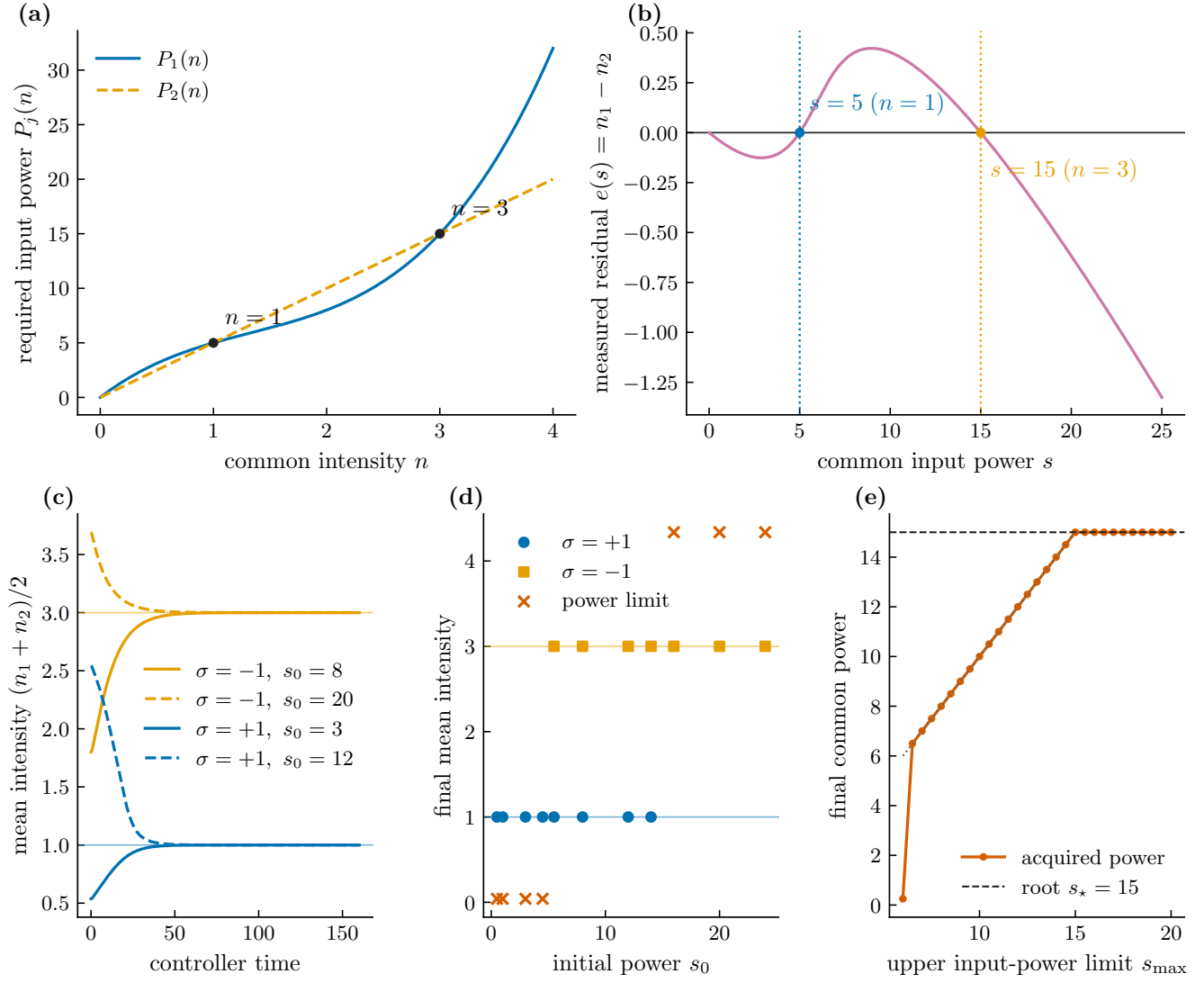


FIG. 4. Two cavities driven by the same field solve the encoded quadratic $n^2 - 4n + 3 = 0$. (a) The cavity input-output laws intersect at $n = 1$ and 3 . (b) Their measured intensity difference vanishes at common powers $s = 5$ and 15 . (c) Representative full complex-amplitude trajectories reach $n = 1$ for one feedback sign and $n = 3$ for the other. (d) Fifteen of 22 runs sampling initial power and feedback sign reach one of those roots with maximum intensity error 2.18×10^{-6} . Runs outside the corresponding basin reach the imposed power limit. (e) Seventeen of 18 upper limits below $s = 15$ exclude the high-power root, whereas all ten limits above it permit acquisition.

The cavity response modifies the magnitude but not the origin of this loss. Implicit differentiation of $P_j[n_j(s)] = s$ gives $\partial_s n_j = 1/P'_j(n_j)$. At an encoded simple root,

$$\chi_{\Delta}^{(\pm)} = -\frac{r_{\pm} q'(r_{\pm})}{P'_1(r_{\pm}) P'_2(r_{\pm})}. \quad (14)$$

Provided both cavity branches remain regular, the feedback rate therefore vanishes linearly with the root separation. This condition is essential because Kerr cavities can also become multivalued [26, 28] or dynamically unstable [27].

All seven simple-root cases in Fig. 7 reach their target,

but the settling time increases from 36.09 to 210.14 across the sampled separations. At exact even multiplicity,

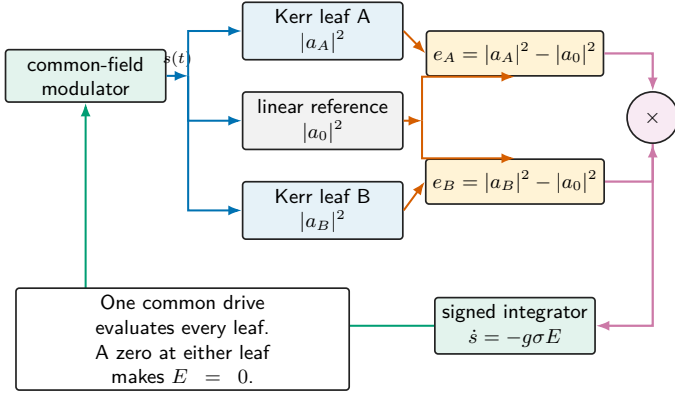
$$q(n) = (n - r)^2 \geq 0,$$

the error has the same sign on both sides. No fixed feedback sign can restore the root from both directions. One side approaches it, while the other is driven to the power limit. The failure is therefore not numerical. A squared error signal contains no first-order indication of which side of the root the controller occupies.

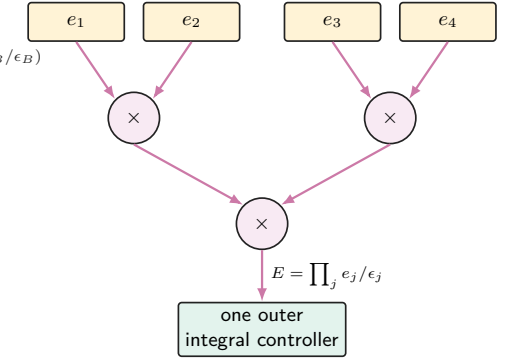
Calibration errors are amplified by the same small slope. Write the measured signal from leaf j as

$$\tilde{e}_j = (1 + \alpha_j)e_j + \beta_j + \xi_j, \quad (15)$$

(a) A shared-reference twinning tree



(b) Quadratic error signals are multiplied



Two quadratic leaves are computed here. Additional product levels multiply the local slope by the normalised response of every inactive leaf.

FIG. 5. The shared-reference construction sends one field through several twinning conditions and returns their product to one controller. (a) One coherent input reaches a linear reference and two Kerr cavities. Balanced detectors form the intensity differences $e_A = |a_A|^2 - |a_0|^2$ and $e_B = |a_B|^2 - |a_0|^2$. A multiplier forms $E = (e_A/\epsilon_A)(e_B/\epsilon_B)$, and one integral controller updates the common input. The product vanishes when either Kerr-cavity intensity equals the reference. (b) Further leaves repeat the same construction. Section VI tests one, two, and three leaves. The diagram does not imply favourable scaling.

where α_j is a gain error, β_j an offset, and ξ_j detector noise. Near a root supplied by leaf k ,

$$\delta s_* \simeq -\frac{\beta_k}{e'_k(s_*)}.$$

The root is therefore most sensitive to offset precisely where it is slowest to acquire. Moreover, an error in a cavity parameter changes the response curve itself. The controller may still find an equality, but that equality need no longer represent the polynomial specified by the nominal parameters.

We test these effects in the full two-leaf cavity model. The simulation includes finite detector bandwidth, detector noise, gain and offset errors, feedback delay, errors in the cavity linewidths, detunings, and Kerr coefficients, and a power actuator with finite range and slew. With parameter errors set to zero, every tested root is retained for noise amplitudes up to 8×10^{-4} and delays up to 0.1 in the dimensionless units of the model. At delay 5, three quarters of the runs retain the intended root. Figure 8(a) records this numerical boundary. It is not an experimental tolerance specification.

Cavity-parameter errors are more damaging. With noise and delay held fixed, the fraction of runs that retain the intended root falls from 1 for the nominal cavities to 0.833 at fractional error 10^{-4} , 0.167 at 3×10^{-4} , and zero at 10^{-3} . Some of these failed runs still reach an equality between the perturbed cavities, as Fig. 8(b) shows. The optical lock is then working, but the encoded equation has changed.

Two distinct requirements follow. The gain, delay, optical relaxation time, and actuator range determine whether the loop can reach and hold an equality [24, 25].

Root separation, the responses of the other leaves, and parameter calibration determine whether that equality remains a well-conditioned solution of the intended equation. An optical root calculation must therefore specify the equation, the accessible power interval, and the calibration of the response curves. A small detector error alone cannot certify all three.

VIII OPTICAL TWINNING AS COMPUTATION

The central operation of this paper is equality under one field. In the original twinning construction, the field is calculated from the states of two materials so that their responses agree [5]. In the feedback construction, the measured disagreement changes the common field until the same equality is reached. For a stationary regular response, the twinning field is a stable fixed point of the loop. When the material states move that field, an integral controller follows with the finite error in Eq. (5). When the optical response has its own dynamics, stability is fixed by the full Jacobian in Eq. (6). These are different regimes of one physical condition, not interchangeable uses of the word root.

The Hubbard calculation makes the distinction concrete. A controller that sees only the current difference can follow the instantaneous twinning field of two interacting rings, but it must lag behind it. Supplying the states allows a feed-forward term to remove most of that lag, because the states contain the missing velocity of the root. Feedback therefore replaces advance knowledge of the field with measurement, acquisition time, and a finite tracking error. That trade is the experimental content of

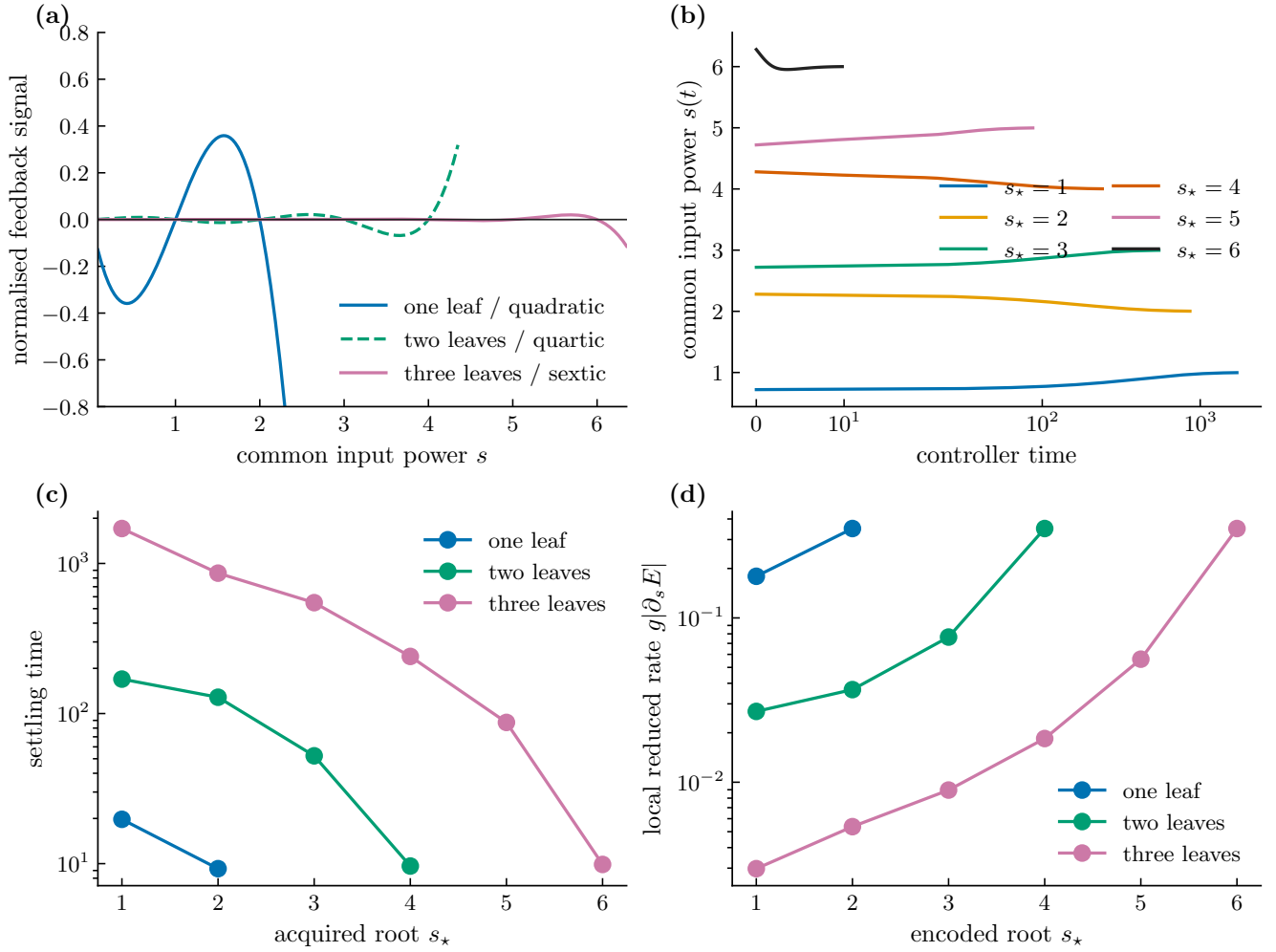


FIG. 6. Composing quadratic twinning conditions gives two, four, and six non-vacuum roots, but progressively less uniform dynamics. (a) Feedback signals for one, two, and three quadratic factors. (b) Full four-cavity trajectories reach all six roots in separately initialised runs with the restoring feedback sign. Each run reaches one root. (c) The slowest settling times increase sharply as factors are added. (d) With three factors, the local convergence rates span a factor of 117.5, because the slope at each root contains the off-root responses of the other leaves [Eq. (13)].

closing the original open-loop construction.

The same loop becomes a computer when the material responses are chosen rather than given. The coefficients of a quadratic set two cavity response curves. Their measured intensity difference evaluates the equation, the common input power is the search variable, and a non-vacuum equality identifies an accessible positive root. Several quadratic conditions can share one reference cavity. Multiplying their error signals gives a factorised quartic with two leaves and a factorised sextic with three. Each run reaches one root according to its initial condition and feedback sign. Repeating the run from several initial conditions recovers the tested root set.

This construction sits at the intersection of several established operations. Balanced detection and signed feedback provide an optical lock [8, 9]. Viewed as a dynamical

system, the same apparatus is a continuous-time root finder. Viewed as an analogue optical circuit, it multiplies signals from nonlinear response elements. Optical media and feedback systems already implement mathematical operators and solve integral or differential equations [10–14]. Photonic reservoirs and recurrent optoelectronic systems instead use their dynamics to process information or seek fixed points [15–18]. The operation introduced here is distinguished by what is set to zero: the difference between two systems driven by the same adjustable field.

The three-leaf calculation also prevents an easy claim of scalability. Four cavities reach six supplied roots, but the local convergence rates span a factor of 117.5. Parameter errors can move the physical equality far enough that the loop returns a root of the wrong response curves. A useful implementation must therefore control the off-root

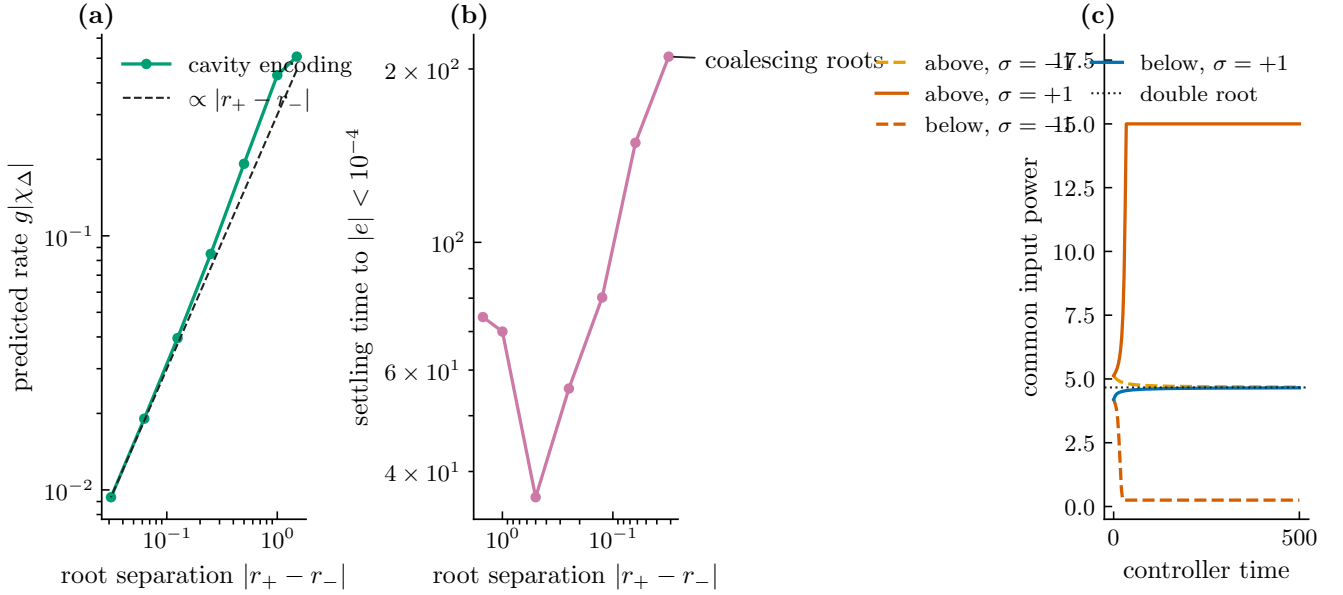


FIG. 7. The conditioning calculation shows the loss of first-order information as two encoded roots coalesce. (a) The predicted local feedback rate vanishes linearly with root separation. (b) All seven simple root cases settle, but the measured settling time grows from 36.09 to 210.14 over the tested separations. (c) At exact repetition each fixed polarity approaches the root from only one side. The opposite-side trajectory reaches the imposed input power limit. The obstruction belongs to the scalar differential response rather than to the choice of gain.

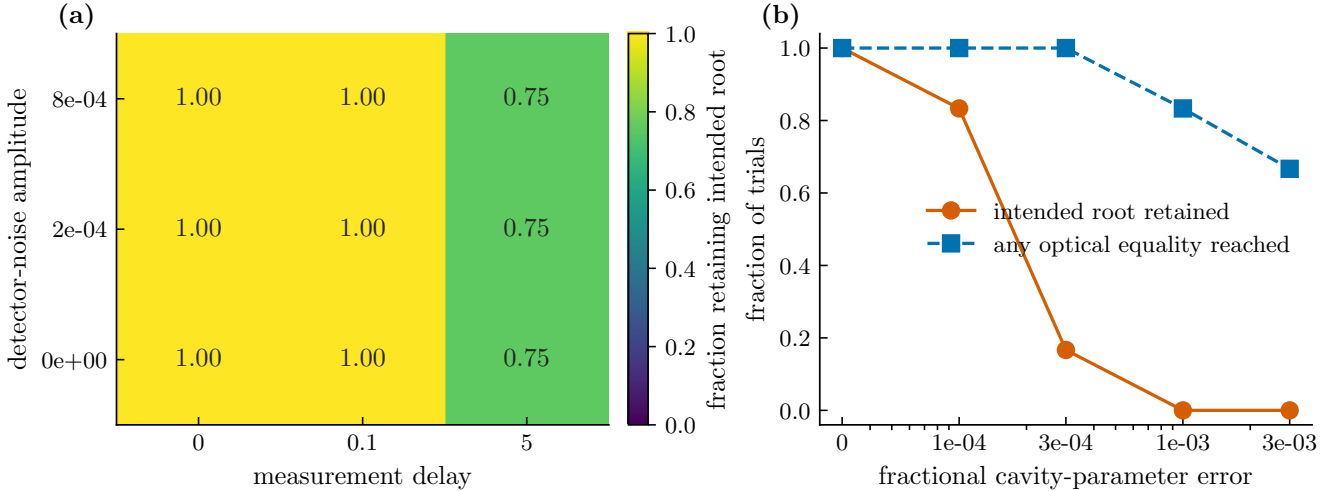


FIG. 8. A cavity loop can remain optically locked after it ceases to solve the intended equation. The simulation includes detector bandwidth, delay, noise, gain and offset errors, cavity-parameter errors, finite actuator range and slew, and anti-windup. (a) Every run retains its intended root over the tested noise range for delays 0 and 0.1. At delay 5, the fraction falls to 0.75. (b) Small cavity-parameter errors can preserve an equality between the perturbed cavities after the encoded root has moved. Fractions include every root and deterministic initial condition in the stated ensemble. The map is numerical evidence, not an experimental tolerance specification.

response of every inactive leaf, the calibration of the encoded coefficients, detector offsets, delay, dynamic range, and the set of initial conditions used to cover the basins. Adding another factor is algebraically simple. Preserving a measurable and correctly calibrated slope at every root

is not.

No experiment is reported here. Driven Kerr cavities, balanced detection, response feedback, and analogue multiplication all have experimental precedents [6, 7, 21, 26, 27, 29], but their assembly into this cir-

cuit remains a proposal. We have also not shown optical advantage, automatic factorisation, arbitrary polynomial encoding, complex roots, or favourable scaling with tree depth. What we have shown is that optical indistinguishability can do more than identify a special drive. With the response laws chosen by supplied coefficients, disagreement supplies the iteration, agreement stops it, and the common field records the solution.

Appendix A: Reduced, moving-root, and augmented linearisations

For a frozen regular root, expand the differential response of Eq. (1) as

$$e(\Phi_\star + \delta\Phi) = \chi_\Delta \delta\Phi + O(\delta\Phi^2).$$

At zero delay, the polarity choice $\sigma\chi_\Delta > 0$ gives $\delta\dot{\Phi} = -\kappa|\chi_\Delta|\delta\Phi$, and hence

$$\delta\Phi(t) = \delta\Phi(0) \exp[-\kappa|\chi_\Delta|t].$$

For $\tau > 0$, substitute $\lambda = i\omega$ into Eq. (4). Separating real and imaginary parts gives

$$\kappa|\chi_\Delta| \cos(\omega\tau) = 0, \quad \omega - \kappa|\chi_\Delta| \sin(\omega\tau) = 0.$$

The first positive crossing has $\omega\tau = \pi/2$ and $\omega = \kappa|\chi_\Delta|$, hence $\kappa|\chi_\Delta|\tau = \pi/2$. The branch connected continuously to the stable root at $\tau = 0$ is stable below this crossing. In the retained scalar sweep, the asymptotic exponent is obtained from a least-squares fit to $\log|e(t)|$ on the stated tail interval. A negative exponent denotes contraction. Finite-time acquisition additionally requires the residual to remain below tolerance for the hold interval.

For a moving root, differentiate $\delta = \Phi - \Phi_\star(t)$ and use the same local expansion:

$$\dot{\delta} = -\kappa\sigma\chi_\Delta\delta - \dot{\Phi}_\star + O(\delta^2).$$

The leading slowly varying solution is $-\dot{\Phi}_\star/(\kappa|\chi_\Delta|)$. With delay, the homogeneous part retains the scalar characteristic equation, while $-\dot{\Phi}_\star$ acts as a forcing term. The state-based comparison adds an independently estimated root velocity to the controller. Because this estimate is formed from successive state inversions, its residual is first-order in the integration step. It tests the missing term in the tracking equation and is not evidence for the detector-only implementation.

Finally, let x contain the real and imaginary components of every cavity amplitude. Linearising $\dot{x} = f(x, s)$, $\dot{s} = -g\sigma E(x)$ gives Eq. (6). On a regular stable optical branch, differentiating $f[x_{ss}(s), s] = 0$ gives

$$\frac{dx_{ss}}{ds} = -A^{-1}b, \quad E'(s_\star) = -c^\top A^{-1}b.$$

The eigenvalue equation for a controller-dominated mode is

$$\lambda = -g\sigma c^\top (\lambda I - A)^{-1}b.$$

Expanding the resolvent about $\lambda = 0$ yields $\lambda = -g\sigma E'(s_\star) + O(g^2/\gamma_{\text{opt}})$. All spectra in Fig. 2 are computed from the real augmented Jacobian by centred finite differences at the exact steady amplitudes.

Appendix B: Cavity mappings and numerical protocols

For the Hubbard calculations, each four-site ring is restricted to two spin-up and two spin-down fermions, giving $\binom{4}{2}^2 = 36$ basis states. Initial states are prepared independently for $U_1 = 0.5$ and $U_2 = 2.0$, then propagated under the common Peierls phase. The tracking ensemble uses material-rate factors 0.5, 1.0, 1.5, gains 24, 36, 46.5, delays 0.004, 0.02, and both the integral-feedback and state-based feed-forward controllers. The representative trace has $g = 36$, $\tau = 0.02$, and unit material-rate factor. The refinement uses $\Delta t = 0.008, 0.004, 0.002$. The instantaneous comparison root is obtained by an independent scalar inversion at every propagated state. It is never supplied to the integral loop.

For the quadratic construction, substituting Eq. (9) into Eq. (8) gives

$$P_1 - P_2 = n[(\Delta_1 - n)^2 + \gamma_1^2 - \gamma_2^2] = n(n^2 + bn + c).$$

The linewidths are positive when $L > 0$ and $L + \Delta_1^2 - c > 0$. The working interval is also chosen so that both $P'_j(n) > 0$, making $n_j(s)$ single valued there. The acquisition runs integrate the full complex amplitudes. The input-power difference is used to design the equality. The controller measures the intensity difference $|a_1|^2 - |a_2|^2$.

For a supplied factor $q_j(s) = s^2 + b_js + c_j$, substitution of the detuning and linewidth in Eq. (11) gives $nK_j^2 q_j(Gn)/G^2$. The depth tests use $G = 5$, $K_j = 1$, and

$$(\gamma_j, \Delta_j) = (\sqrt{4.99}, 0.3), (\sqrt{4.99}, 0.7), (\sqrt{4.99}, 1.1)$$

for the three leaves. These encode the supplied consecutive factors with roots $1, \dots, 6$. Each detector scale ϵ_j is the largest absolute steady residual of that leaf on the stated depth-specific operating interval. A single gain is chosen at each depth from the largest local terminal slope and then held fixed across its roots. The acquisition horizon is set from the independently predicted local rate, capped at 3000, so a slow but stable root is not reclassified as a structural failure. Depth-three targets are repeated at $\Delta t = 0.02, 0.01, 0.005$.

The full-loop spectra are calculated at the exact steady amplitudes. For each real state coordinate x_k , the centred finite-difference step is $2 \times 10^{-6} \max(1, |x_k|)$. The quadratic Jacobian has five real variables and the two-leaf Jacobian seven. The reported dominant eigenvalue is the eigenvalue with largest real part. Stability requires that real part to be negative. The reduced comparison uses a separate centred derivative of the steady intensity residual with respect to common power.

The robustness ensemble integrates the two-leaf full-cavity loop with $\Delta t = 0.02$ to time 420. The detector time constant is 0.04, the actuator slew limit 0.8, and the power range is $[0.08, 6.35]$ with anti-windup. Detector-noise amplitudes are $0, 2 \times 10^{-4}, 8 \times 10^{-4}$. Delays are 0, 0.1, 5, and each map cell contains every root and two deterministic seeds. The separate parameter-error ensemble uses fractional cavity errors $0, 10^{-4}, 3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}$, three deterministic seeds per root, detector noise 2×10^{-4} , and delay 0.04. An intended root is retained when the tail mean power lies within 0.03 and the RMS power jitter is below 0.02. Equality lock is recorded separately from that classification.

Each simulation writes its results to CSV tables and a JSON summary before any figure is made. The plotting

routines read only those tables. Producer, data, and figure hashes are retained with the parameter sweeps and the explicit $n \leq 8$ many-body bound. The figures can therefore be regenerated without rerunning the simulations.

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- [1] G. McCaul, C. Orthodoxou, K. Jacobs, G. H. Booth, and D. I. Bondar, Driven imposters: Controlling expectations in many-body systems, [Physical Review Letters](#) **124**, 183201 (2020), arXiv:1911.05006.
 - [2] G. McCaul, C. Orthodoxou, K. Jacobs, G. H. Booth, and D. I. Bondar, Controlling arbitrary observables in correlated many-body systems, [Physical Review A](#) **10.1103/PhysRevA.101.053408** (2020), arXiv:1912.06173.
 - [3] A. B. Magann, G. McCaul, H. A. Rabitz, and D. I. Bondar, Sequential optical response suppression for chemical mixture characterization, [Quantum](#) **6**, 626 (2022), arXiv:2010.13859.
 - [4] G. McCaul, A. F. King, and D. I. Bondar, Non-uniqueness of non-linear optical response, [Annalen der Physik](#) **10.1002/andp.202100523** (2022), arXiv:2110.06189.
 - [5] G. McCaul, A. F. King, and D. I. Bondar, Optical indistinguishability via twinning fields, [Physical Review Letters](#) **10.1103/PhysRevLett.127.113201** (2021), arXiv:2103.01162.
 - [6] J. Masur, D. I. Bondar, and G. McCaul, Dynamical generation of epsilon-near-zero behaviour via tracking and feedback control, arXiv preprint [10.48550/arXiv.2301.12069](#) (2023), arXiv:2301.12069.
 - [7] V. Bilokon, E. Bilokon, and D. I. Bondar, All you need is amplifier: Spectral imposters without pulse shaping, arXiv preprint [10.48550/arXiv.2603.05417](#) (2026), arXiv:2603.05417.
 - [8] T. W. Hänsch and B. Couillaud, Laser frequency stabilization by polarization spectroscopy of a reflecting reference cavity, [Optics Communications](#) **35**, 441 (1980).
 - [9] E. D. Black, An introduction to pound–drever–hall laser frequency stabilization, [American Journal of Physics](#) **69**, 79 (2001).
 - [10] A. Silva, F. Monticone, G. Castaldi, V. Galdi, A. Alù, and N. Engheta, Performing mathematical operations with metamaterials, [Science](#) **343**, 160 (2014).
 - [11] N. M. Estakhri, B. Edwards, and N. Engheta, Inverse-designed metastructures that solve equations, [Science](#) **363**, 1333 (2019).
 - [12] J. Cederquist and S. H. Lee, Coherent optical feedback for the analog solution of partial differential equations, [Journal of the Optical Society of America](#) **70**, 944 (1980).
 - [13] J. Hou, J. Dong, and X. Zhang, Optical solver for a system of ordinary differential equations based on an external feedback assisted microring resonator, [Optics Letters](#) **42**, 2310 (2017).
 - [14] A. Cordaro, B. Edwards, V. Nikkhah, A. Alù, N. Engheta, and A. Polman, Solving integral equations in free space with inverse-designed ultrathin optical metagratings, [Nature Nanotechnology](#) **18**, 365 (2023), arXiv:2202.05219.
 - [15] L. Appeltant, M. C. Soriano, G. Van der Sande, J. Danckaert, S. Massar, J. Dambre, B. Schrauwen, C. R. Mirasso, and I. Fischer, Information processing using a single dynamical node as complex system, [Nature Communications](#) **2**, 468 (2011).
 - [16] K. Vandoorne, P. Mechet, T. Van Vaerenbergh, M. Fiers, G. Morthier, D. Verstraeten, B. Schrauwen, J. Dambre, and P. Bienstman, Experimental demonstration of reservoir computing on a silicon photonics chip, [Nature Communications](#) **5**, 3541 (2014).
 - [17] F. Böhm, G. Verschaffelt, and G. Van der Sande, A poor man’s coherent ising machine based on opto-electronic feedback systems for solving optimization problems, [Nature Communications](#) **10**, 3538 (2019).
 - [18] K. P. Kalinin, J. Gladrow, J. Chu, *et al.*, Analog optical computer for AI inference and combinatorial optimization, [Nature](#) **645**, 354 (2025).
 - [19] G. McCaul, K. Jacobs, and D. I. Bondar, Towards single atom computing via high harmonic generation, [The European Physical Journal Plus](#) **10.1140/epjp/s13360-023-03649-3** (2023), arXiv:2104.06322.
 - [20] G. McCaul, J. S. Toterogongora, W. Otieno, S. Savell’ev, A. Zagorskin, and A. G. Balanov, Minimal quantum reservoirs with hamiltonian encoding, [Chaos: An Interdisciplinary Journal of Nonlinear Science](#) **10.1063/5.0282921** (2025), arXiv:2505.22575.
 - [21] T. Fujita, Y. Toba, Y. Miyoshi, and M. Ohashi, Optical analog multiplier based on phase sensitive amplification, in [2013 18th OptoElectronics and Communications Conference Held Jointly with 2013 International Conference on Photonics in Switching](#) (Optica Publishing Group, 2013) p. TuPS_8.
 - [22] M. Yildirim, N. U. Dinc, I. Oguz, D. Psaltis, and

- C. Moser, Nonlinear processing with linear optics, *Nature Photonics* **18**, 1076 (2024).
- [23] T. Wu, Y. Li, L. Ge, and L. Feng, Field-programmable photonic nonlinearity, *Nature Photonics* **19**, 725 (2025).
- [24] K. Pyragas, Continuous control of chaos by self-controlling feedback, *Physics Letters A* **170**, 421 (1992).
- [25] N. D. Hayes, Roots of the transcendental equation associated with a certain difference-differential equation, *Journal of the London Mathematical Society* **s1-25**, 226 (1950).
- [26] H. M. Gibbs, S. L. McCall, and T. N. C. Venkatesan, Differential gain and bistability using a sodium-filled fabry-perot interferometer, *Physical Review Letters* **36**, 1135 (1976).
- [27] P. D. Drummond and D. F. Walls, Quantum theory of optical bistability. i. nonlinear polarisability model, *Journal of Physics A: Mathematical and General* **13**, 725 (1980).
- [28] K. Ikeda, Multiple-valued stationary state and its instability of the transmitted light by a ring cavity system, *Optics Communications* **30**, 257 (1979).
- [29] L. A. Lugiato and R. Lefever, Spatial dissipative structures in passive optical systems, *Physical Review Letters* **58**, 2209 (1987).